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Movies & Entertainment Ontology

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Movies & Entertainment Ontology

M&E Ontology Requirements Specification Document

1 Purpose

The purpose of building this Ontology is to provide a consensual knowledge model of the different types of Movies & Entertainment domain to be used by public.

2 Scope

The ontology has to focus just on representing structured knowledge related to the movie and entertainment domain, including movies, people involved in production, and user interactions. The level of granularity is directly related to the competency questions and terms identified.

3 Implementation Language

The ontology has to be implemented in OWL language.

4 Intended End-Users

CV - Casual Viewers / Movie Enthusiasts: General public interested in browsing, discovering, and exploring movie and entertainment content.

RA - Researchers & Academics: Scholars studying media, genre patterns, cultural content distribution, or industry trends.

IP- Industry Professionals: Directors, producers, writers, and editors referencing structured domain knowledge.

PD - Platform Developers / Recommender Systems: Engineers building recommendation engines, streaming search, or metadata-driven applications.

5 Intended Uses

Use 1. To look for movies options.

Use 2. Gather information about specific movie genre.

Use 3. Gathering structured information about cast, crew, and production details.

Use 4. Analysing user feedback (ratings and reviews) to assess content quality.

Use 5. Supporting research on entertainment industry trends, awards, and distribution platforms.

6 Ontology Requirements

a. Non-Functional Requirements

NFR1. The ontology must be consistent should be able to answer all the competency questions.

NFR2. The ontology must satisfy FAIR (Findable, Accessible, Interoperable, Reusable) principle.

b. Functional Requirements: Groups of Competency Questions

Group 1: Media Content

- CQ1. What are all the movies available in the system? (CV, RA, PD)
- CQ2. Which movies belong to a specific genre? (CV, RA, PD)
- CQ3. Which movies are produced by a specific production company? (CV, IP)
- CQ4. Which movies are created in a specific language? (CV, RA)
- CQ5. Which movies belong to a specific country? (CV, IP)
- CQ6. Which movies have a specific certification (e.g., PG-13)? (CV, PD)
- CQ7. Which movies have won awards? (CV, RA, IP)
- CQ8. Which movies are available on a specific online retailer? (CV, PD)
- CQ9. Which movies are associated with a particular director? (CV, IP)
- CQ10. Which movies belong to multiple genres? (CV, PD)

Group 2: Genre

- CQ1. What genres exist in the system? (CV, RA)
- CQ2. Which movies belong to the Action genre? (CV, PD)
- CQ3. Which movies belong to the Drama genre? (CV, RA)
- CQ4. Which movies belong to more than one genre? (RA, PD)
- CQ5. What is the genre of a specific movie? (CV)
- CQ6. Which genre has the highest number of movies? (RA, PD)
- CQ7. Which genres are associated with a specific director? (RA, IP)
- CQ8. Which movies belong to both Comedy and Romance? (CV, RA)
- CQ9. Which genres are most popular among movies? (RA, PD)
- CQ10. Which genres are linked to award-winning movies? (RA, IP)

Group 3: Cast & Crew (Person)

- CQ1. Who are the actors in a specific movie? (CV, IP)
- CQ2. Which movies did a specific actor perform in? (CV, IP)
- CQ3. Who is the director of a specific movie? (CV, IP)
- CQ4. Which movies are produced by a specific producer? (RA, IP)
- CQ5. Which writers contributed to a specific movie? (IP)
- CQ6. Which editors worked on a specific movie? (IP)
- CQ7. Which musical artists contributed to a movie? (IP)
- CQ8. Which persons are involved in a specific movie? (CV, IP)
- CQ9. Which actors worked with a specific director? (RA, IP)
- CQ10. Which persons contributed to multiple movies? (RA, IP)

Group 4: User Feedback (Rating & Review)

- CQ1. What is the rating of a specific movie? (CV, PD)
- CQ2. Which movies have the highest ratings? (CV, RA, PD)
- CQ3. Which users gave ratings to a specific movie? (PD)
- CQ4. What reviews are written for a specific movie? (CV)
- CQ5. Which user wrote a review for a movie? (PD)
- CQ6. Which movies have the most reviews? (RA, PD)
- CQ7. What is the average rating of a movie? (RA, PD)
- CQ8. Which movies have low ratings? (CV, PD)
- CQ9. Which movies have both high ratings and many reviews? (RA, PD)
- CQ10. Which users give the highest ratings? (PD)

Group 6: Users

- CQ1. Who are the users of the system? (PD)
- CQ2. Which users have rated a specific movie? (PD)
- CQ3. Which users have written reviews? (PD)
- CQ4. Which user has given the most ratings? (PD)
- CQ5. Which user reviewed multiple movies? (PD)
- CQ6. What activities are performed by a specific user? (PD)
- CQ7. Which users interact with a specific genre of movies? (RA, PD)
- CQ8. Which users have rated multiple genres? (RA, PD)
- CQ9. Which users are most active in the system? (PD)
- CQ10. Which users have given both ratings and reviews? (PD)

Group 7: Production & Awards

- CQ1. Which production companies exist in the system? (RA, IP)
- CQ2. Which movies are produced by a specific company? (RA, IP)
- CQ3. Which movies have won awards? (CV, RA, IP)
- CQ4. Which awards are associated with a specific movie? (CV, IP)
- CQ5. Which actors have received awards? (RA, IP)
- CQ6. Which directors have won awards? (RA, IP)
- CQ7. Which movies have both awards and high ratings? (RA, PD)
- CQ8. Which production companies produced award-winning movies? (RA, IP)
- CQ9. Which awards are given to specific genres? (RA)
- CQ10. Which movies have both certification and awards? (RA, IP)

Group 8: Distribution Platforms (OnlineRetailer)

- CQ1. Which online retailers exist in the system? (CV, PD)
- CQ2. Which movies are available on a specific retailer? (CV, PD)
- CQ3. Which movies are available on multiple retailers? (PD)
- CQ4. Which platforms provide movies of a specific genre? (CV, PD)
- CQ5. Which platform has the highest-rated movies? (RA, PD)
- CQ6. Which movies are distributed by a specific retailer? (PD)
- CQ7. Which platforms provide movies in a specific language? (CV, PD)
- CQ8. Which platforms host award-winning movies? (CV, RA)
- CQ9. Which retailers provide movies from a specific country? (RA, PD)
- CQ10. Which platforms have the most diverse movie collection? (RA, PD)

7 Pre-Glossary of Terms

Terms from Competency Questions

Media	User
Movie	Rating
TV Series	Reviews
Season	Production Company
Actor	Award
Producer	Online Retailor
Director	Language
Writer	Country